

Project Design Phase-II

Data Flow Diagram & User Stories Document

Date:	30 September 2026
Team ID:	SWTID-2026-1128
Project Title:	Auto Ticket Classification using Flow Designer
Maximum Marks:	4 Marks

1. Project Overview & Problem Statement

The school IT helpdesk receives multiple support requests daily from students, teachers, and administrative staff related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues. Currently, IT staff manually review each ticket to identify the issue type, select the correct category and subcategory, and inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the volume of requests increases. To overcome these challenges, the project implements an Automated Ticket Classification system utilizing ServiceNow Flow Designer and Natural Language Understanding / Automation workflows to streamline ticket ingestion, automatic categorization, routing, and instant stakeholder notifications.

2. Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD depicts the system requirements graphically, showing how data enters and leaves the system, what processes transform the information, and where data is stored.

2.1 Context Level Diagram (DFD Level 0)

The Context Diagram outlines the system boundary between external entities and the Automated Ticket Classification System:

- **External Entities:** Students, Teachers, IT Staff, and System Administrators.
- **Inputs:** Support Request details (Issue description, user credentials, category preferences, ticket updates).
- **Outputs:** Ticket status confirmations, automated categorization logs, assignment notifications, and resolution reports.

2.2 Detailed Data Flow (DFD Level 1 Process Breakdown)

- **1.0 Ticket Ingestion & Validation:** Captures user issues via web portal or email, validates requester identity against the User Directory, and creates an initial raw incident record.

- **2.0 Automated Text Processing & NLU:** Parses the ticket summary and description, extracts key intent keywords, and interfaces with the Classification Engine to evaluate category matching.
- **3.0 Flow Designer Logic & Categorization:** Applies predefined business rules and flow triggers to assign accurate Category, Subcategory, and Priority level.
- **4.0 Incident Assignment & Routing:** Queries the Assignment Matrix in the Data Store to route the classified ticket to the designated IT Support Group.
- **5.0 Notification & Status Dispatch:** Triggers automatic email/SMS alerts to the caller confirming ticket creation and notifies assigned IT technicians.

3. Requirement Specifications & User Stories

The following matrix details the functional user stories and acceptance criteria for the Auto Ticket Classification system across different user roles (Students, Teachers, IT Staff/Executives, and Administrators).

User Type	Functional Req. (Epic)	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
Student / Teacher	Ticket Submission	USN-1	As a user, I can submit a support ticket describing my technical issue so that the IT department can assist me.	1. Ticket form accepts title & description. 2. Confirmation receipt is immediately shown on portal.	High	Sprint-1
Student / Teacher	Automated Confirmation	USN-2	As a user, I receive an instant automated notification once my ticket is submitted, confirming the assigned category and ticket number.	1. Email notification contains ticket ID, category, and priority. 2. Link provided to track status.	High	Sprint-1
Student / Teacher	Self-Service Tracking	USN-3	As a user, I can view the real-time status and updates of my submitted support requests on my personal dashboard.	1. Dashboard shows open & closed tickets. 2. Status updates refresh in real-time.	Medium	Sprint-2
IT Support Executive	Auto-Categorization	USN-4	As an IT technician, I want incoming tickets to be automatically categorized and prioritized by Flow Designer so I don't have to triage them manually.	1. Category, Subcategory, and Impact are populated on creation. 2. Accuracy rate meets defined flow thresholds.	High	Sprint-1
IT Support Executive	Smart Queue Assignment	USN-5	As an IT technician, I want tickets automatically routed to my team's queue based on issue type (e.g. Network, Hardware, Account).	1. Ticket appears in correct support queue. 2. Team members receive assignment notification.	High	Sprint-1
IT Support Executive	Manual Category Override	USN-6	As an IT technician, I want the ability to manually override ticket categories if auto-classification was incorrect, feeding feedback	1. IT staff can update Category/Subcategory dropdowns. 2. Change audit log captures original vs	Medium	Sprint-2

User Type	Functional Req. (Epic)	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
			into the flow.	corrected fields.		
Administrator	Flow Designer Rules	USN-7	As an Administrator, I can configure and edit classification rules and triggers in Flow Designer without writing complex code.	1. Admin UI allows editing keyword triggers. 2. Flow updates take effect immediately upon publishing.	High	Sprint-1
Administrator	Reporting & Analytics	USN-8	As an Administrator, I can generate analytics reports on ticket volume, classification accuracy, and average resolution times.	1. Analytics dashboard displays accuracy metrics. 2. Export options available for PDF/Excel reports.	Medium	Sprint-2